

The Real Number *System*

SCORE 3.0 Basic Understanding

1.1 Converting Rational Numbers

NC.8.NS.1 Understand that every number has a decimal expansion. Building upon the definition of a rational number, know that an irrational number is defined as a non-repeating, non-terminating decimal.

Convert each number as directed. Write fractions in **simplest form** and show your work.

1.

Write as a decimal: $\frac{3}{8}$

= 0.375

2.

Write as a fraction: 0.25

= $\frac{1}{4}$

1.2 Real Number Classification

NC.8.NS.1 Understand that every number has a decimal expansion. Building upon the definition of a rational number, know that an irrational number is defined as a non-repeating, non-terminating decimal.

Classify the number as **Rational (R)** or **Irrational (IR)**.

3.

Classify: $\sqrt{144}$

Classification: **R**

SCORE 4.0 Target Content

1.1 Converting Rational Numbers

NC.8.NS.1 Understand that every number has a decimal expansion. Building upon the definition of a rational number, know that an irrational number is defined as a non-repeating, non-terminating decimal.

Convert each number as directed. Write fractions in **simplest form** and show your work.

4.

Write as a fraction: $0.\overline{6}$

= $\frac{2}{3}$

5.

Write as a decimal: $2\frac{5}{6}$

= 2.833...

1.2 Real Number Classification

NC.8.NS.1 Understand that every number has a decimal expansion. Building upon the definition of a rational number, know that an irrational number is defined as a non-repeating, non-terminating decimal.

Classify the number as **Rational (R)** or **Irrational (IR)**.

6.

Classify: $\sqrt{20}$

Classification: **IR**

Now simplify each expression first, *then* classify the result.

7.

Classify: $\sqrt{3} \times \sqrt{27}$

8.

Classify: $\sqrt{32} \div \sqrt{2}$

Classification: **R**

Classification: **R**

1.3 Estimating Non-Perfect Square Roots

NC.8.NS.2 Use rational approximations of irrational numbers to compare the size of irrational numbers and locate them approximately on a number line. Estimate the value of expressions involving: square roots and cube roots to the tenths; π to the hundredths.

Estimate each value to the **nearest tenth**, and plot it on the number line.

9.

Estimate: $\sqrt{40}$

$\approx$ **6.3**

10.

Estimate: $\sqrt{95}$

$\approx$ **9.7**

SCORE 5.0 Advanced Content

1.2 Real Number Classification

NC.8.NS.1 Understand that every number has a decimal expansion. Building upon the definition of a rational number, know that an irrational number is defined as a non-repeating, non-terminating decimal.

Evaluate or simplify each expression if possible, *then* classify the result as **Rational (R)** or **Irrational (IR)**.

11.

Classify: $-(\sqrt{49} - \sqrt{36})$

Classification: **R**

12.

Classify: $3 + \sqrt{5}$

Classification: **IR**

1.4 Ordering Real Numbers

NC.8.NS.2 Use rational approximations of irrational numbers to compare the size of irrational numbers and locate them approximately on a number line.

Order the set from **least to greatest**.

13.

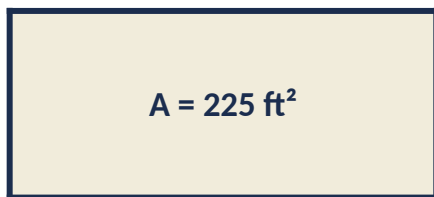
Set: { 0.6, $\sqrt{2}$, 1.5, $5/3$ }

Order = **0.6, $\sqrt{2}$, 1.5, $5/3$**

1.5 Area

A square garden has an area of 225 ft². Find the side length and the perimeter. Show all work.

14.



Side length = **15 ft**

Perimeter = **60 ft**